Chapter B

Formula sheet with explanations

Subject	Formula	Check to carry out
Arithmetic sequence	$u_n = u_0 + nr$	Is the initial index 0 or 1?
Arithmetic sum	$N\frac{\text{first} \quad + \text{last}}{2}$	Count the terms: last index $-$ first index $+1$.
Geometric sequence	$u_n = u_0 q^n$	q is it a multiplier and not a percentage ?
Geometric sum	$u_0 \dfrac{1-q^{n+1}}{1-q}$	$q \neq 1$ and $n+1$ terms.
Average rate	$q = Q^{\frac{1}{n}}, t = q - 1$	Do not divide the overall rate by n.
Logarithm	$a^x = b \iff x = \dfrac{\log b}{\log a}$	$a > 0, a \neq 1, b > 0$.
Reciprocal function	$\left(\dfrac{1}{x}\right)' = -\dfrac{1}{x^2}$	Exclude $x = 0$.
Linear fit	$y = ax + b$	Estimate, observed range, unit.
Conditional probability	$P_A(B) = \dfrac{P(A \cap B)}{P(A)}$	$P(A) > 0$ and order of conditioning.
Law of total probability	$P(B) = \sum_i P(A_i) P_{A_i}(B)$	Do A_i form a partition?
Binomial distribution	$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$	Independence, same p, $0 \leq k \leq n$.
Binomial expectation	$E(X) = np$	Long-run mean.